7. Integration Techniques

7.1. Integration by Parts

7.1.3. Definite Integrals

7.1b Learning Objectives

- I can use integration by parts to solve definite integrals.
- I understand when and how to use integration by parts.
- I can choose an appropriate u and v.
- I can use integration by parts to reduce integrals.

Recall the product rule once more to see what we do when the integrals have bounds.

Product Rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Integrating both sides from a to b, and rearranging slightly, we end up with the following

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}[f(x)g(x)] &= f(x)g'(x) + f'(x)g(x) \\ f(x)g(x)|_a^b &= \int_a^b f(x)g'(x)\,\mathrm{d}x + \int_a^b f'(x)g(x)\,\mathrm{d}x \\ f(x)g(x)|_a^b - \int_a^b f'(x)g(x)\,\mathrm{d}x &= \int_a^b f(x)g'(x)\,\mathrm{d}x \end{aligned}$$

Relabeling the last line with a few substitutions $u = f(x) \Rightarrow \mathrm{d}u = f'(x)\,\mathrm{d}x$ and $v = g(x) \Rightarrow \mathrm{d}v = g'(x)\,\mathrm{d}x$ we end up with

Theorem 7.1.1 Integration by Parts for Definite Integrals

$$\int_a^b u\,\mathrm{d}v = [uv]_a^b - \int_a^b v\,\mathrm{d}u$$

Note

Note that in practice, we still have the same variable of integration before and after applying integration by parts. We aren't really substituting anything since the variables u and v don't stick around. This way, the bounds are still for the original variable of integration and shouldn't be changed. This is different from the substitution method, where the variable of integration does change, so the bounds must also change.

Example 7.1.1

Find

$$\int_0^1 x3^x\,\mathrm{d}x.$$

Solution

$$u = x, \quad \mathrm{d}v = 3^x \, \mathrm{d}x,$$
$$\mathrm{d}u = \mathrm{d}x, \quad v = \int 3^x \, \mathrm{d}x$$
$$= \frac{1}{\ln(3)} 3^x$$

Then we have

$$\begin{aligned}
\int_0^1 x 3^x \, \mathrm{d}x &= \left[\frac{x}{\ln(3)} 3^x\right]_0^1 - \int_0^1 \frac{1}{\ln(3)} 3^x \, \mathrm{d}x \\
&= \left[\frac{1}{\ln(3)} 3 - 0\right] - \left[\frac{1}{(\ln(3))^2} 3^x\right]_0^1 \\
&= \frac{3}{\ln(3)} - \left[\frac{3}{(\ln(3))^2} - \frac{1}{(\ln(3))^2}\right] \\
&= \frac{3}{\ln(3)} - \frac{2}{(\ln(3))^2}
\end{aligned}$$

Example 7.1.2

Find

$$\int_0^1 \arctan(x) \, \mathrm{d}x.$$

Solution

We want to pick $u = \arctan(x)$ since inverse trig functions get simpler (for integrating) after we take the derivative. Notice that this just leaves $\mathrm{d}v = \mathrm{d}x$. That is okay, and can sometimes be helpful even!

$$u = \arctan(x), \; \mathrm{d}v = \mathrm{d}x,$$
$$\mathrm{d}u = \frac{1}{1 + x^2} \, \mathrm{d}x, \quad v = x$$

Then

$$\begin{aligned}
\int_0^1 \arctan(x) \, \mathrm{d}x &= [x \arctan(x)]_0^1 - \int_0^1 \frac{x}{1 + x^2} \, \mathrm{d}x \\
&= \frac{\pi}{4} - 0 - \int_0^1 \frac{x}{1 + x^2} \, \mathrm{d}x
\end{aligned}$$

Now we just need to evaluate

$$\int_0^1 \frac{x}{1 + x^2} \, \mathrm{d}x$$

Take a second to finish this on your own before checking the solution below. You know how to do this.

We need a substitution, since we see a function and its derivative: Let $w = 1 + x^2$. Then $\mathrm{d}w = 2x\,\mathrm{d}x$. Since this is an actual substitution, we need to change the bounds as well $w(0) = 1$ and $w(1) = 2$. Lastly, if you want to call this u instead of w, that is fine too. I just picked something to avoid confusion with integration by parts. Then

$$\begin{aligned}\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x &= \frac{1}{2}\int_1^2 \frac{1}{w}\,\mathrm{d}w \\ &= \frac{1}{2}[\ln|w|]_1^2 \\ &= \frac{1}{2}\ln(2)\end{aligned}$$

Altogether

$$\int_0^1 \arctan(x)\,\mathrm{d}x = \frac{\pi}{4} - \frac{1}{2}\ln(2).$$

Exercise 7.1.1

Find

$$\int_3^5 \ln(x)\,\mathrm{d}x.$$

Sometimes you have to use integration by parts multiple times to solve an integral. This happens with powers of trig functions, “double transendental functions”, and some other things.

Example 7.1.3

Find

$$\int e^x \cos(x)\,\mathrm{d}x.$$

When thinking about what to pick for u here, if we pick either e^x or $\cos(x)$ we run into similar problems: nothing ever really changes.

We can take advantage of the fact that nothing really changes by some really slick algebra shown in the following solution.

Solution

Let’s give the thing we want to find a name: Let $I = \int e^x \cos(x)\,\mathrm{d}x$.

Now, apply integration by parts a few times (picking the same type of thing for u) until you see I show up again, like so

$$\begin{aligned} u &= \cos(x), & \mathrm{d}v &= e^x\,\mathrm{d}x, \\ \mathrm{d}u &= -\sin(x)\,\mathrm{d}x, & v &= e^x \end{aligned}$$

$$
\begin{aligned}
I &= \int e^x \cos(x)\,\mathrm{d}x \\
&= \cos(x)e^x - \underbrace{\int e^x(-\sin(x))\,\mathrm{d}x}_{\neq I}
\end{aligned}
$$

Since the thing in the integral is not I, we repeat on this new integral term.

$$
\begin{aligned}
u &= -\sin(x), & \mathrm{d}v &= e^x\,\mathrm{d}x, \\
\mathrm{d}u &= -\cos(x)\,\mathrm{d}x, & v &= e^x
\end{aligned}
$$

Then

$$
\begin{aligned}
I &= \int e^x \cos(x)\,\mathrm{d}x \\
&= \cos(x)e^x - \left[\int e^x(-\sin(x))\,\mathrm{d}x\right] \\
&= \cos(x)e^x - \left[-\sin(x)e^x - \int e^x(-\cos(x))\,\mathrm{d}x\right] \\
&= \cos(x)e^x + \sin(x)e^x - \underbrace{\int e^x \cos(x)\,\mathrm{d}x}_{=I}
\end{aligned}
$$

So we end up with the algebraic equation which we can solve for the answer, I:

$$
\begin{aligned}
I &= \cos(x)e^x + \sin(x)e^x - I \\
2I &= \cos(x)e^x + \sin(x)e^x \\
I &= \frac{1}{2}[\cos(x)e^x + \sin(x)e^x]
\end{aligned}
$$

Exercise 7.1.2

Given $n \geq 2$ is an integer, prove the **reduction formula**:

$$
\int (\sin(x))^n\,\mathrm{d}x = -\frac{1}{n}\cos(x)\sin^{n-1}(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,\mathrm{d}x
$$

Alternatively, use integration by parts to find $\int \sin^2(x)\,\mathrm{d}x$. Hint use $u = \sin(x)$ for the first integration by parts step and follow similar steps as the previous problem.

If you want, we can go over this in office hours.

7.1b Section Summary:

- We learned how to handle bounds on integrals when we need to use integration by parts.
- We also looked at some examples that need multiple steps of integration by parts to solve.